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EE1030/EE2100: Matrix Theory

Lecture 23: The Fundamental Subspaces of a Matrix

Topics :

1. Column Space and Null Space of a Matrix
 2. Row Space and Left Null Space of a Matrix
 3. Existence and Uniqueness of Solutions to Linear Systems
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Systems of Linear Equations and Matrices

given a system of linear Equations

Questions of interest

a) do the system
of linear Equations
have a solution?

$$\left\{ \begin{array}{l} a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n = b_m \end{array} \right\}$$

m Equations &

n unknowns $(x_1, \dots, x_n)$

Yes No

unique infinitely many.

Can be Rep as $A\vec{x} = \vec{b}$ where

$$\vec{b} = [b_1, \dots, b_m]^T \in \mathbb{R}^m$$

$$\vec{x} = [x_1, \dots, x_n]^T \in \mathbb{R}^n$$

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n} = [\vec{a}_1, \dots, \vec{a}_n]$$

where $\vec{a}_i \in \mathbb{R}^m$

① way to assess the nature of
solutions from row/column vectors of A?

② Algorithm to determine nature & the
Corresponding sols?

$$\text{or } A = \begin{bmatrix} \vec{b}_1^T \\ \vdots \\ \vec{b}_m^T \end{bmatrix} \text{ where } \vec{b}_i \in \mathbb{R}^n$$

The Column Space and Null Space of a Matrix - Rank and Nullity

$A = [\vec{a}_1 \dots \vec{a}_n]$ where $\vec{a}_i \in \mathbb{R}^m$
 → Coefficient matrix of system of linear eqn's.

Col. Space ($\text{Col}(A)$) → Set of all possible linear combinations of column vectors of A
 $= \text{Span}(\vec{a}_1, \dots, \vec{a}_n) \rightarrow$
 $\text{Col}(A)$ is a subspace of $\mathbb{R}^m$ & $(\vec{a}_1, \dots, \vec{a}_n)$ is the spanning set
 Can they form a basis? → Yes if $\vec{a}_1, \dots, \vec{a}_n$ are linearly independent
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 Can happen if $m \geq n$

Rank of $A \rightarrow \dim(\text{Col}(A)) \rightarrow$ no. of linearly independent column vectors
 ↳ Rank $\in [1, \min(m, n)]$ (For a Non-zero Matrix)

Null space of A : ($\text{Null}(A)$): → $\text{Null}(A) = \{ \vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0} \} \rightarrow \vec{0} \in \text{Null}(A)$
 $\vec{x} \neq \vec{0}$ exists if the column vectors of A are linearly dependent

Nullity → $\dim(\text{Null}(A))$

The Row Space and Left Null Space of a Matrix

Row space of a matrix $\rightarrow$ Set of all possible linear comb of row vectors of A .

$$\text{Span}(\vec{b}_1, \dots, \vec{b}_m)$$

Column space of A^T ; $\text{Col}(A^T) \rightarrow$ Subspace of $\mathbb{R}^n$

dim of Row space
= no. of linearly independent rows

is it true? $\dim(\text{Col}(A^T)) = \dim(\text{Col}(A))$

Left Null space: $\text{Null}(A^T) = \{ \vec{y} \in \mathbb{R}^m : A^T \vec{y} = \vec{0} \}$

Existence and Uniqueness of Solutions to Linear Systems

① Solving $A\vec{x} = \vec{b}$ $\rightarrow$ finding coefficients of linear combination α_i s.t.

$$\sum_{i=1}^n \alpha_i \vec{a}_i = \vec{b}$$

and $\vec{x} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$

② Sol. exists if $\vec{b}$ is in the span $(\vec{a}_1, \dots, \vec{a}_n) \Rightarrow \vec{b} \in \text{Col}(A)$

③ Say $\exists \vec{x}_c \neq \vec{0}$ s.t. $A\vec{x}_c = \vec{0}$; $\vec{x}_c \in \text{Null}(A)$ }
 $\exists \vec{x}_p \neq \vec{0}$ s.t. $A\vec{x}_p = \vec{b}$

with $\vec{x}_p + \alpha \vec{x}_c$ be a solution to $A\vec{x} = \vec{b} \rightarrow A(\vec{x}_p + \alpha \vec{x}_c) = A\vec{x}_p + \alpha A\vec{x}_c$
 $= \vec{b} + \vec{0}$
 $= \vec{b}$

↓
Yes.

If $\dim(\text{Null}(A)) > 0$, then $A\vec{x} = \vec{b}$ will have no solution or infinitely many solutions.